A COUNTEREXAMPLE TO THE ERDŐS–MAHLER CONJECTURE

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Abstract. We disprove a 1939 conjecture of Erdős and Mahler by constructing, for every real $\tau \geq 4$, a Cantor set of numbers of irrationality exponent τ, each having infinitely many regular continued-fraction convergents of the form $2^a/3^b$.

1. Introduction

For an irrational real number ξ, its *irrationality exponent* $\mu(\xi)$ is the supremum of the real numbers $\nu > 0$ for which $0 < |\xi - u/v| < v^{-\nu}$ holds for infinitely many reduced fractions u/v with $v \geq 1$. Thus ξ is a Liouville number precisely when $\mu(\xi) = \infty$. Write p_n/q_n for the regular continued-fraction convergents of ξ, and $\mathrm{P}(m)$ for the largest prime factor of $|m|$ when m is a nonzero integer, with $\mathrm{P}(\pm 1) = 1$.

In 1939, Erdős and Mahler [3] investigated the prime factors of continued-fraction numerators and denominators. They proved that if $\mathrm{P}(q_{n-1}q_nq_{n+1})$ is bounded for infinitely many n, then ξ is a Liouville number. They conjectured that the same conclusion should hold if $\mathrm{P}(p_nq_n)$ is bounded for infinitely many n. The question was subsequently studied by Fraenkel [6] and recalled by Erdős in his reminiscences of Mahler [4]. In 2004, Bugeaud [1, Problem 43, p. 215] included the conjecture among the open problems on continued fractions.

Conjecture (Erdős–Mahler)**.** If $\mathrm{P}(p_nq_n)$ is bounded for infinitely many n, then ξ is a Liouville number.

We disprove the conjecture even when the selected numerators are powers of 2 and the corresponding denominators are powers of 3.

Theorem. *For every real $\tau \geq 4$, there exists a Cantor set $K_\tau \subset (1, 2)$ such that every $\xi \in K_\tau$ has irrationality exponent τ and infinitely many regular continued-fraction convergents of the form $2^a/3^b$, with a, b positive integers.*

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By Ridout's theorem [11], bounded prime factors of the denominators of infinitely many convergents already imply transcendence. Fraenkel [5, 6] studied rational approximation with prescribed prime factors and constructed Liouville numbers with infinitely many convergents subject to such restrictions. Fraenkel and Borosh [7] subsequently obtained non-Liouville examples by allowing additional cofactors in the numerators and denominators. When both cofactors are bounded, their Hausdorff-dimension formula gives zero. Lelis and Marques [8, Theorem 1.1] proved the conjectured conclusion under an additional spacing condition on the indices of the selected convergents.

More recently, Bugeaud and Nguyen [2, Question 4.1, p. 32] revisited the conjecture and constructed further examples with restricted prime factors, leaving open whether the numbers they constructed are Liouville. This illustrates the main difficulty: prescribing infinitely many convergents does not exclude exceptionally good rational approximations at other denominators.

The key ingredient is a counting estimate for the prescribed fractions that admit exceptionally close rational approximations. The estimate is independent of the intervals used in the construction and, together with the distribution of an irrational rotation, ensures that sufficiently many choices remain at each stage. This allows us to construct Cantor sets with the prescribed irrationality exponent.

The paper is organized as follows. Section 2 states the counting proposition and deduces the main theorem from it. Section 3 proves the proposition, using rotation estimates, the p-adic Subspace Theorem, and multiplicative relations among the prescribed fractions.

2. The counting proposition and the main theorem

We first state the counting estimate underlying the construction and deduce the main theorem from it. The proof of the estimate is given in the next section.

Notation. All logarithms are natural, and $\mathbb{N} = \{1, 2, \dots\}$. Rational fractions have positive denominators. We write $|I|$ for the length of an interval or arc and $|A|$ for the cardinality of a finite set. The symbols O and $\ll$ have their usual meaning; implied constants are absolute unless a dependence is indicated.

Set

$$\alpha = \frac{\log 3}{\log 2}, \qquad a(b) = \lceil b\alpha \rceil, \qquad R(b) = \frac{2^{a(b)}}{3^b} \qquad (b \in \mathbb{N}).$$

Unique factorization shows that α is irrational, so $0 < a(b) - b\alpha < 1$. Consequently, $R(b)$ is the unique reduced fraction in $(1, 2)$ with denominator 3^b and numerator a power of 2. Let $\mathcal{D}$ be the set of denominators of the regular continued-fraction convergents of α. For $x \in \mathbb{R}$ write $\{x\} = x - \lfloor x \rfloor$, and define

$$\Phi(y) = 1 - \frac{\log y}{\log 2} \qquad (1 \le y \le 2).$$

Then $\Phi(R(b)) = \{b\alpha\}$ and $|\Phi'(y)| = (y \log 2)^{-1}$.

For $4 < \nu \leq 5$, a reduced fraction $u/v \in [1, 2]$ is *exceptional for b at exponent ν* if $u/v \neq R(b)$ and

$$\left| R(b) - \frac{u}{v} \right| < \frac{1}{16 v^{\nu}}. \tag{1}$$

For $N \in \mathbb{N}$ and $D > 0$, let $E_N(D, \nu)$ be the number of integers $b \in [N, 2N)$ admitting an exceptional fraction with $v > D$.

Proposition 2.1 (Exceptional approximations)**.** *There exists an absolute constant $C > 0$ such that, for every $4 < \nu \leq 5$ and $D \geq 2$,*

$$\limsup_{\substack{N \to \infty \\ N \in \mathcal{D}}} \frac{E_N(D, \nu)}{N} \leq C D^{2-\nu}.$$

The deduction of the theorem also uses an elementary estimate for the distribution of the fractions $R(b)$, obtained from the convergents of α.

Lemma 2.2 (Rotation estimate)**.** *For every arc $I \subset \mathbb{R}/\mathbb{Z}$ and every $N \in \mathcal{D}$,*

$$\#\{b \in \mathbb{Z} : N \leq b < 2N,\ \{b\alpha\} \in I\} = N|I| + O(1), \tag{2}$$

with an absolute implied constant.

Proof. Choose a convergent p/N of α. Since $\gcd(p, N) = 1$, the points

$$N\alpha + \frac{ip}{N} \pmod 1, \qquad 0 \leq i < N,$$

form a translate of the uniform grid of spacing $1/N$. Thus $N|I| + O(1)$ of these points lie in I. By the convergent estimate $|\alpha - p/N| < N^{-2}$ [1, Corollary 1.4, p. 10],

$$\left| (N + i)\alpha - \left(N\alpha + \frac{ip}{N} \right) \right| = i \left| \alpha - \frac{p}{N} \right| < \frac{1}{N}.$$

The two counts can therefore differ only at grid points within distance $1/N$ of an endpoint of I. There are $O(1)$ such points, uniformly in I and N, proving (2). □

We turn to the construction. The rotation estimate and Proposition 2.1 ensure that two admissible choices remain at each stage.

Fix $\tau \geq 4$. For $t \in \mathbb{N}$, put $Q = 3^t$ and define

$$J_t = (R(t) + Q^{-\tau},\ R(t) + 2Q^{-\tau}),$$
$$K(t) = [R(t) + Q^{-\tau},\ R(t) + 3Q^{-\tau}].$$

Since $R(t) \leq 2 - Q^{-1}$, both intervals lie in $(1, 2)$ for sufficiently large t. Set

$$D_t = \frac{Q^{\tau - 1}}{6}, \qquad \nu_t = 4 + \frac{1}{t}.$$

An index b is *admissible at t* if $R(b) \in J_t$ and no fraction exceptional for b at exponent ν_t has denominator greater than D_t.

Lemma 2.3 (Admissible choices)**.** *For each $\tau \geq 4$, there exists t_0 such that, for every integer $t \geq t_0$, the interval $[N, 2N)$ contains at least two distinct indices admissible at t whenever $N \in \mathcal{D}$ is sufficiently large, depending on t.*

Proof. Fix t sufficiently large that $J_t \subset (1, 2)$ and $D_t \geq 2$, and put $\rho_t = |\Phi(J_t)|$. Since

$$\frac{1}{2\log 2} \leq |\Phi'(y)| \leq \frac{1}{\log 2} \qquad (1 \leq y \leq 2),$$

the mean value theorem gives

$$\frac{Q^{-\tau}}{2\log 2} \leq \rho_t \leq \frac{Q^{-\tau}}{\log 2}.$$

By the rotation estimate (2), there are $N\rho_t + O(1)$ indices $b \in [N, 2N)$ for which $R(b) \in J_t$. Of these, at most $E_N(D_t, \nu_t)$ are inadmissible. If $A_t(N)$ denotes the number of admissible indices, Proposition 2.1 therefore yields

$$\liminf_{\substack{N\to\infty \\ N\in\mathcal{D}}} \frac{A_t(N)}{N} \geq \rho_t - CD_t^{2-\nu_t} \geq \rho_t - 36CQ^{-2(\tau-1)},$$

where C is the absolute constant in that proposition and we used $\nu_t > 4$.

The error relative to ρ_t satisfies

$$\frac{36CQ^{-2(\tau-1)}}{\rho_t} \leq 72C(\log 2)Q^{2-\tau} \longrightarrow 0 \qquad (t \to \infty),$$

since $\tau \geq 4$. Hence, for all sufficiently large t, the preceding lower limit is at least $\rho_t/2 > 0$. With such a t fixed, $A_t(N)$ tends to infinity as $N \to \infty$ through $\mathcal{D}$. In particular, there are at least two admissible indices for every sufficiently large $N \in \mathcal{D}$. □

Proof of the theorem. Choose t_0 sufficiently large for Lemma 2.3. For each $t \geq t_0$, take $N \in \mathcal{D}$ with $N > Q = 3^t$, large enough to provide two admissible indices $b < c$ in $[N, 2N)$. We first show that $K(b)$ and $K(c)$ are disjoint and lie in the interior of $K(t)$.

For $d = b, c$, admissibility gives $R(d) \in J_t$. Moreover, $d > t$, so $3^d \geq 2Q$. The left endpoint of $K(d)$ is greater than $R(t) + Q^{-\tau}$, whereas its right endpoint satisfies

$$\begin{aligned} R(d) + 3 \cdot 3^{-\tau d} &< R(t) + (2 + 3 \cdot 2^{-\tau})Q^{-\tau} \\ &< R(t) + 3Q^{-\tau}, \end{aligned}$$

since $\tau \geq 4$. Thus

$$K(b), K(c) \subset \operatorname{int} K(t).$$

To prove disjointness, observe that $c < 2b$. The distinct reduced fractions $R(b)$ and $R(c)$ can both be expressed with denominator 3^c. Consequently,

$$|R(b) - R(c)| \geq 3^{-c} > 3^{-2b} > 3 \cdot 3^{-\tau b}.$$

Each of $K(b)$ and $K(c)$ lies to the right of its defining fraction, at distance at most $3 \cdot 3^{-\tau b}$. Hence the two intervals are disjoint.

Fix such a pair of successors for every $t \geq t_0$ and iterate the construction, labeling the two choices by 0 and 1. Each infinite binary sequence determines a nested chain

$$K(t_0) \supset K(t_1) \supset K(t_2) \supset \cdots .$$

Along this chain, put $r_j = R(t_j)$ and $Q_j = 3^{t_j}$. Since $Q_{j+1} \geq 2Q_j$, the intervals satisfy

$$|K(t_j)| = 2Q_j^{-\tau} \leq 2Q_0^{-\tau} 2^{-\tau j}.$$

Their intersection therefore consists of a unique point ξ. The uniform decay of the interval lengths makes the resulting map from $\{0,1\}^{\mathbb{N}}$ continuous, while disjointness of the successors makes it injective. By compactness, this map is a homeomorphism onto its image, a Cantor set $K_\tau \subset (1,2)$.

Fix $\xi \in K_\tau$ and consider its associated chain. Strict nesting gives

$$Q_j^{-\tau} < \xi - r_j < 3Q_j^{-\tau} \qquad (j \geq 0). \tag{3}$$

We first show that ξ is irrational. If $\xi = u/v$ were rational, then $\xi \neq r_j$ and rational separation would give

$$|\xi - r_j| \geq \frac{1}{vQ_j}.$$

Since $Q_j \to \infty$ and $\tau \geq 4$, this contradicts (3) for large j.

Moreover, for every j,

$$0 < \xi - r_j < 3Q_j^{-\tau} < \frac{1}{2Q_j^2}.$$

As each r_j is reduced, Legendre's criterion [1, Theorem 1.8, p. 10] shows that r_j is a regular continued-fraction convergent of ξ. These convergents have numerators that are powers of 2 and strictly increasing denominators that are powers of 3. For every $0 < s < \tau$, (3) gives

$$0 < \xi - r_j < Q_j^{-s}$$

for all sufficiently large j. Hence $\mu(\xi) \geq \tau$.

We now prove the reverse inequality. Let $u/v \in [1,2]$ be reduced with $v \geq Q_0$, and choose the unique j such that

$$Q := Q_j \leq v < Q_{j+1} =: S.$$

If $u/v = r_j$, then $v = Q$ and (3) gives $|\xi - u/v| > v^{-\tau}$. We may therefore assume $u/v \neq r_j$ and distinguish two ranges of denominators.

(i) $v \leq D_{t_j}$. Since r_j is reduced with denominator Q, rational separation and (3) give

$$\begin{aligned}\left|\xi - \frac{u}{v}\right| &\geq \frac{1}{vQ} - \frac{3}{Q^\tau} \\ &\geq \frac{1}{2vQ} \geq \frac{1}{2v^2},\end{aligned}$$

where the second inequality follows from $6v \leq Q^{\tau-1}$ and the last from $Q \leq v$.

(ii) $v > D_{t_j}$. Since $v < S$ and r_{j+1} is reduced with denominator S, we have $u/v \neq r_{j+1}$ and

$$\left| r_{j+1} - \frac{u}{v} \right| \geq \frac{1}{Sv}.$$

On the other hand, $S \geq 3$ and $v < S$ imply $6v < S^3$. As $\tau \geq 4$, (3) gives

$$\begin{aligned} |\xi - r_{j+1}| &< \frac{3}{S^\tau} \leq \frac{3}{S^4} < \frac{1}{2Sv} \\ &\leq \frac{1}{2} \left| r_{j+1} - \frac{u}{v} \right|. \end{aligned}$$

The index t_{j+1} is admissible at t_j. Since $v > D_{t_j}$, the definition of admissibility yields

$$\left| r_{j+1} - \frac{u}{v} \right| \geq \frac{1}{16 v^{\nu_{t_j}}}.$$

Consequently,

$$\left| \xi - \frac{u}{v} \right| \geq \frac{1}{2} \left| r_{j+1} - \frac{u}{v} \right| \geq \frac{1}{32 v^{\nu_{t_j}}}.$$

Combining the two ranges, we have

$$\left| \xi - \frac{u}{v} \right| \geq \frac{1}{32 v^{\nu_{t_j}}} \qquad (Q_j \leq v < Q_{j+1},\ u/v \neq r_j).$$

Since $t_j \to \infty$, we have $\nu_{t_j} \to 4$. Given $\varepsilon > 0$, it follows that $\nu_{t_j} \leq 4 + \varepsilon/2$ for all sufficiently large j. Moreover, $v^{\varepsilon/2} \geq 32$ for all sufficiently large v. Thus

$$\left| \xi - \frac{u}{v} \right| \geq v^{-4-\varepsilon} \geq v^{-\tau-\varepsilon}$$

for every sufficiently large denominator v and every reduced $u/v \in [1, 2]$ distinct from the prescribed convergent in its range.

For the prescribed convergents, (3) gives $|\xi - r_j| > Q_j^{-\tau}$. Finally, rationals outside $[1, 2]$ are bounded away from ξ, while only finitely many reduced fractions in $[1, 2]$ have bounded denominator. It follows that $\mu(\xi) \leq \tau$. Together with the lower bound, this proves

$$\mu(\xi) = \tau \qquad \text{for every } \xi \in K_\tau.$$

□

3. Proof of the counting proposition

We begin with a direct estimate for small exceptional denominators and a height bound for the remaining ones. We then count the exceptional fractions in fixed denominator ranges by exploiting the multiplicative structure of $R(b)$, and sum the resulting estimates.

3.1. **Small denominators and a height bound.** The rotation estimate gives a direct count for exceptional fractions with denominator at most $N^{1/3}$. The p-adic Subspace Theorem will provide the height bound needed for larger denominators.

Lemma 3.1 (Small-denominator approximations)**.** *There is an absolute constant $C > 0$ such that, for every $4 < \nu \le 5$, $D \ge 2$, and $N \in \mathcal{D}$,*

$$E_N(D,\nu) \le E_N(N^{1/3},\nu) + C\big(ND^{2-\nu} + N^{2/3}\big). \tag{4}$$

Proof. If $D \ge N^{1/3}$, the assertion follows from the monotonicity of $E_N(D,\nu)$ in D. Suppose therefore that $D < N^{1/3}$.

For $V \ge 2$, there are $O(V^2)$ reduced fractions $u/v \in [1,2]$ with $V < v \le 2V$, since each denominator v admits at most $v+1$ numerators. As $|\Phi'(y)| \le (\log 2)^{-1}$ on $[1,2]$, the image under Φ of the $(16v^\nu)^{-1}$-neighborhood of any such fraction, intersected with $[1,2]$, is an interval of length $O(V^{-\nu})$. By $\Phi(R(b)) = \{b\alpha\}$ and (2), this fraction can be exceptional for at most $O(NV^{-\nu}+1)$ indices $b \in [N,2N)$.

Thus denominators in $(V,2V]$ contribute at most $O(NV^{2-\nu}+V^2)$ indices. Cover $(D,N^{1/3}]$ by the intervals $(V,2V]$, where $V = 2^iD$ and $i \ge 0$, truncating the last one if necessary. Their total contribution is at most a constant times

$$\sum_{\substack{i\ge 0\\ 2^iD<N^{1/3}}} \big(N(2^iD)^{2-\nu} + (2^iD)^2\big) \ll ND^{2-\nu} + N^{2/3}.$$

Indeed, the first sum is geometric with ratio $2^{2-\nu} \le 1/4$, whereas the second is dominated by its largest term. The implied constants are absolute for $4 < \nu \le 5$. Adding the indices admitting exceptional denominators greater than $N^{1/3}$ gives (4). □

By Lemma 3.1, the proof of Proposition 2.1 reduces to showing that, for each fixed $4 < \nu \le 5$,

$$E_N(N^{1/3},\nu) = o(N) \qquad (N \to \infty,\ N \in \mathcal{D}).$$

For distinct reduced fractions $R(b)$ and u/v, rational separation gives

$$\left|R(b) - \frac{u}{v}\right| \ge \frac{1}{3^b v}.$$

Together with (1), this yields $16v^{\nu-1} < 3^b$, and hence only $\log v = O(N)$ when $N \le b < 2N$. The summation over larger denominator ranges requires a bound $\log v = o(N)$, uniform over the exceptional fractions in each block.

Fix $4 < \nu \le 5$, and let W_N be the largest value of $\log v$ among fractions exceptional for some $b \in [N,2N)$, with $W_N = 0$ if there are none. The elementary bound just obtained ensures that W_N is finite.

Lemma 3.2 (Height bound)**.** *For every $\varepsilon > 0$ there exists $b_0(\varepsilon)$ such that, for every $b \geq b_0(\varepsilon)$ and every reduced fraction $u/v \in [1,2]$ distinct from $R(b)$,*

$$\left| R(b) - \frac{u}{v} \right| \geq 3^{-\varepsilon b} v^{-2}.$$

Consequently, for each fixed $4 < \nu \leq 5$,

$$W_N = o(N) \qquad (N \to \infty,\ N \in \mathcal{D}). \tag{5}$$

Proof. Put $P = 2^{a(b)}$ and $S = 3^b$, so that $S < P < 2S$. For a reduced fraction $u/v \in [1,2]$ distinct from P/S, set

$$d = \gcd(Pv, Su), \qquad U = \frac{Pv}{d}, \qquad W = \frac{Su}{d}, \qquad e = |Pv - Su|.$$

Then $e \geq 1$ and $(U, W) = 1$. Every prime divisor of d is 2 or 3; otherwise it would divide both u and v. For a positive integer n, write $[n]_6 = 2^{v_2(n)} 3^{v_3(n)}$, with v_p the p-adic valuation. Normalize the p-adic absolute values by $|p|_p = p^{-1}$. Since d has no prime divisors outside $\{2, 3\}$,

$$|UW|_2 |UW|_3 = [UW]_6^{-1} = \frac{d^2}{PS[uv]_6}.$$

Take the linear forms $X - Y, Y$ at the real place and X, Y at 2 and 3. Their product at (U, W) satisfies

$$\Pi := |(U - W)W|\, |UW|_2 |UW|_3 = \frac{eu}{P[uv]_6} \leq 2v^2 \left| \frac{P}{S} - \frac{u}{v} \right|,$$

where we used $e = Sv|P/S - u/v|$, $u \leq 2v$, and $S < P$.

Suppose that the first assertion fails for arbitrarily large b. For the corresponding fractions, rational separation gives

$$\frac{1}{Sv} \leq \left| \frac{P}{S} - \frac{u}{v} \right| < S^{-\varepsilon} v^{-2},$$

so $v < S$. Hence, with $H = \max(U, W)$, we have $H \leq 2Sv < 2S^2$, and therefore

$$\Pi < 2S^{-\varepsilon} \leq 2^{1+\varepsilon/2} H^{-\varepsilon/2}.$$

Let $\delta = \min(\varepsilon/4, 1/2) > 0$. For sufficiently large H, the last bound is smaller than $(U^2 + W^2)^{-\delta/2}$. The pairs (U, W) are primitive, and the two forms at each place are linearly independent with determinant one. The p-adic Subspace Theorem [12, p. 246, (1.3)–(1.5)] therefore places all such pairs of sufficiently large height on finitely many rational lines through the origin. Each line contains at most one positive primitive pair, and there are only finitely many pairs of bounded height. Thus only finitely many pairs (U, W) can arise.

On the other hand,

$$0 < \left| \frac{U}{W} - 1 \right| = \frac{|P/S - u/v|}{u/v} < S^{-\varepsilon} v^{-2} \longrightarrow 0 \qquad (b \to \infty),$$

which is impossible for a finite set of positive primitive pairs. This proves the first assertion.

For N sufficiently large in terms of ε, the first assertion and (1) give $16v^{\nu-2} < 3^{\varepsilon b}$ for every fraction exceptional at exponent ν with $N \le b < 2N$. Since $b < 2N$,

$$\limsup_{\substack{N\to\infty \\ N\in\mathcal{D}}} \frac{W_N}{N} \le \frac{2\varepsilon \log 3}{\nu - 2}.$$

Letting $\varepsilon \to 0$ proves (5). □

3.2. **Products in a fixed denominator range.** The key observation is that the products $R(b)R(c)$, with $b, c \in [N, 2N)$, take only $O(N)$ distinct values. We show that this property persists for sufficiently close rational approximants whose denominators lie in a suitable range.

Fix $4 < \nu \le 5$, $1 < \theta < \nu/4$, and $N \in \mathcal{D}$. For $w > 0$, put

$$V = e^{w/\theta}, \qquad H = e^w = V^\theta.$$

Let $\mathcal{A}(w)$ be the set of indices $b \in [N, 2N)$ admitting an exceptional fraction at exponent ν with denominator

$$\max\{V, N^{1/3}\} < v \le H.$$

The restriction $4\theta < \nu$ ensures that the approximation errors are smaller than the separation between distinct rational products.

Lemma 3.3 (Products of exceptional fractions)**.** *For each $b \in \mathcal{A}(w)$, choose an exceptional fraction $s_b = u_b/v_b$ with $\max\{V, N^{1/3}\} < v_b \le H$. Put*

$$X = \{s_b : b \in \mathcal{A}(w)\}, \qquad M = |X|.$$

Then

$$|\mathcal{A}(w)| \ll M, \tag{6}$$

with an absolute implied constant. Moreover, the product set $XX = \{ss' : s, s' \in X\}$ satisfies

$$|XX| \le 4N. \tag{7}$$

Proof. We first bound the number of indices assigned to a single fraction $s = u/v \in X$. By (1), the values $\Phi(R(b))$ corresponding to these indices lie in an interval of length $O(v^{-\nu})$ about $\Phi(s)$, since $|\Phi'| \le (\log 2)^{-1}$ on $[1, 2]$. The rotation estimate (2) therefore gives at most

$$O(Nv^{-\nu} + 1) = O(1)$$

such indices, where we used $v > N^{1/3}$ and $\nu > 4$. This proves (6).

For $b, c \in [N, 2N)$, write $\ell = b + c$. Then

$$R(b)R(c) = 2^{a(b)+a(c)}3^{-\ell}.$$

Since $0 < a(d) - d\alpha < 1$ for every $d \in \mathbb{N}$, we have

$$\alpha\ell < a(b) + a(c) < \alpha\ell + 2.$$

Thus, for fixed ℓ, the exponent $a(b)+a(c)$ has at most two possible values. As $2N \le \ell \le 4N-2$, the products $R(b)R(c)$ take at most $4N$ distinct values.

We transfer this bound to the products of elements of X. For $b, c \in \mathcal{A}(w)$, all four fractions lie in $[1,2]$, and hence

$$\begin{aligned} |s_b s_c - R(b)R(c)| &\le 2|s_b - R(b)| + 2|s_c - R(c)| \\ &< \frac{1}{4}V^{-\nu}. \end{aligned}$$

Consequently, if $R(b)R(c) = R(b')R(c')$, then

$$|s_b s_c - s_{b'} s_{c'}| < \frac{1}{2}V^{-\nu}.$$

Each rational product has reduced denominator at most H^2. Distinct such products differ by at least H^{-4}. Since

$$V^{-\nu} < V^{-4\theta} = H^{-4},$$

the corresponding rational products must be equal. Thus every value of $R(b)R(c)$ determines at most one value of $s_b s_c$, proving (7). □

3.3. **Classes modulo rational powers.** We use (7) to bound the number of classes occupied by X modulo rational powers. A consequence of the Plünnecke–Ruzsa inequality controls the number of classes, while a lower bound for linear forms in logarithms and a 3-adic valuation bound their sizes.

Lemma 3.4 (Number of classes)**.** *Let A be a finite nonempty subset of a torsion-free abelian group G, written additively, and put $L = |A+A|/|A|$. For every $q = 2^k$, $k \ge 1$, the image of A in G/qG has at most L^{3k} elements.*

Proof. For finite subsets $A, B \subset G$, write $A \pm B = \{a \pm b : a \in A,\ b \in B\}$ and $m \cdot A = \{ma : a \in A\}$. Choose one representative $a \in A$ from each class in the image of A in G/qG. The sets $a - q \cdot A$ belong to distinct cosets of qG and are therefore disjoint. Since dilation by q is injective, each contains $|A|$ elements. Thus the number of classes is at most

$$\frac{|A - q \cdot A|}{|A|}.$$

The Plünnecke–Ruzsa inequality [10, Theorem 1.2] gives

$$|A - 2 \cdot A| \le |A - A - A| \le L^3 |A|.$$

To iterate this bound, we use Ruzsa's triangle inequality

$$|U - W|\,|V| \le |U - V|\,|V - W|$$

for finite nonempty sets [10, proof of Theorem 1.2, (3)]. Taking $U = A$, $V = 2^{i-1} \cdot A$, and $W = 2^i \cdot A$, and using injectivity of dilation, we obtain

$$\frac{|A - 2^i \cdot A|}{|A|} \le \frac{|A - 2^{i-1} \cdot A|}{|A|} \frac{|A - 2 \cdot A|}{|A|} \qquad (i \ge 2).$$

Induction yields $|A - 2^k \cdot A| \le L^{3k}|A|$, as required. □

For a positive rational number r/s in lowest terms, write

$$h(r/s) = \log \max\{r, s\}.$$

Two positive rationals belong to the same class modulo qth powers if their quotient is a qth power in $\mathbb{Q}_{>0}$.

Lemma 3.5 (Size of a class)**.** *Let $4 < \nu \le 5$, $1 < \theta < \nu/4$, $N \in \mathcal{D}$, and $w \ge 1$. Let X be as in Lemma 3.3, with one representative index chosen for each element of X. Put $x = \log(N/w)$, and let q be the least power of two with $q \ge x^2$.*

There exists an absolute constant $x_0 > 0$ such that, whenever $x \ge x_0$, any two elements $s_b, s_c \in X$ in the same class modulo rational qth powers satisfy

$$\frac{s_c}{s_b} = \frac{R(c)}{R(b)}, \qquad |c - b| \log 3 \le 2w + \log 2.$$

Consequently, each class contains $O(w)$ elements of X, with an absolute implied constant.

Proof. It suffices to consider distinct elements s_b, s_c in the same class. Write $s_c/s_b = g^q$, where $g \in \mathbb{Q}_{>0} \setminus \{1\}$. Since each selected fraction has denominator at most $H = e^w$ and numerator at most $2H$, the reduced numerator and denominator of s_c/s_b are at most $2H^2$. Hence

$$qh(g) = h(s_c/s_b) \le 2w + \log 2 \le 3w. \tag{8}$$

In particular, the height of g is reduced by a factor q.

We first prove that the rational-power relation is an exact identity between the corresponding prescribed fractions. Consider

$$\Lambda = (a(c) - a(b)) \log 2 - (c - b) \log 3 - q \log g.$$

Since $s_c/s_b = g^q$, we have

$$\Lambda = \log \frac{R(c)}{s_c} - \log \frac{R(b)}{s_b}.$$

The logarithm has derivative at most one on $[1, 2]$. Applying (1) at b and c therefore gives

$$|\Lambda| < \frac{1}{8} V^{-\nu} = \frac{1}{8} \exp(-\nu w/\theta). \tag{9}$$

Suppose that $\Lambda \ne 0$. We apply Matveev's lower bound [9, Corollary 2.3, (2.6), with (1.3)]. For positive rational bases $\gamma_i \ne 1$ and integer coefficients m_i, $1 \le i \le 3$, choose

$$A_i \ge \max\{h(\gamma_i), |\log \gamma_i|, 0.16\}, \qquad B = \max\left\{1, \max_{1 \le i \le 3} \frac{|m_i| A_i}{A_3}\right\}.$$

For a nonzero linear form in their logarithms, the cited bound gives

$$\log \left| \sum_{i=1}^{3} m_i \log \gamma_i \right| > -C_0 A_1 A_2 A_3 (1 + \log B),$$

where C_0 is absolute. The result does not require multiplicative independence of the bases.

Apply this estimate with

$$(\gamma_1, \gamma_2, \gamma_3) = (2, 3, g), \qquad (A_1, A_2, A_3) = (\log 2, \log 3, 3w/q).$$

These height parameters are admissible: $|\log g| \le h(g)$, while (8) and $g \neq 1$ give

$$\log 2 \le h(g) \le 3w/q.$$

Moreover, $b, c \in [N, 2N)$ imply

$$|c - b| < N, \qquad |a(c) - a(b)| \le \alpha N + 1.$$

Thus the coefficients of $\log 2$ and $\log 3$ are $O(N)$, whereas that of $\log g$ has absolute value q. Since $N/w = e^x$, the weighted definition of B yields

$$B \ll \frac{qN}{w}, \qquad 1 + \log B \ll 1 + \log q + x.$$

As $x^2 \le q < 2x^2$, Matveev's bound now gives

$$-\log|\Lambda| \ll \frac{w}{q}(1 + \log q + x) \ll \frac{w}{x},$$

with absolute implied constants.

On the other hand, (9) gives

$$-\log|\Lambda| > \frac{\nu w}{\theta} + \log 8 > 4w.$$

For x sufficiently large, these estimates are incompatible. Hence $\Lambda = 0$, and therefore

$$\frac{s_c}{s_b} = 2^{a(c)-a(b)}3^{b-c} = \frac{R(c)}{R(b)}.$$

Taking 3-adic valuations, we obtain $v_3(s_c/s_b) = b - c$. Since $|v_3(z)| \log 3 \le h(z)$ for every positive rational z, (8) gives

$$|c - b| \log 3 \le h(s_c/s_b) \le 2w + \log 2.$$

Fixing one representative index in a class, all other representative indices therefore lie in an interval of length $O(w)$. They are distinct integers, so the class contains $O(w)$ elements. The same bound holds for a singleton, since $w \ge 1$. □

Lemma 3.6 (Count in a denominator range)**.** *There exist absolute constants $c, C, x_0 > 0$ such that, for every $4 < \nu \le 5$, $1 < \theta < \nu/4$, $N \in \mathcal{D}$, and $w \ge 1$ with $x = \log(N/w) \ge x_0$,*

$$\frac{|\mathcal{A}(w)|}{N} \le C \exp\left(-\frac{cx}{\log(2 + x)}\right).$$

Proof. We may assume $\mathcal{A}(w) \neq \varnothing$. Take $x_0 \ge 2$ large enough for Lemma 3.5, and let $q = 2^k$ be the least power of two with $q \ge x^2$.

By Lemma 3.4, applied to X in the multiplicative group $\mathbb{Q}_{>0}$, the number of classes modulo rational qth powers meeting X is at most

$$\left(\frac{|XX|}{M}\right)^{3k} \le \left(\frac{4N}{M}\right)^{3k},$$

where we used (7). Lemma 3.5 bounds the number of elements in each class by $C_1 w$, with C_1 absolute. Consequently,

$$M \le C_1 w \left(\frac{4N}{M}\right)^{3k}.$$

Putting $y = M/(4N)$ and using $w/N = e^{-x}$, we obtain

$$y^{3k+1} \le \frac{C_1}{4} e^{-x}.$$

For sufficiently large x, the right-hand side is at most $e^{-x/2}$. Moreover, $q < 2x^2$ gives

$$3k + 1 \ll \log(2 + x),$$

with an absolute implied constant. Taking the $(3k+1)$st root therefore yields

$$\frac{M}{N} \ll \exp\left(-\frac{cx}{\log(2+x)}\right)$$

for some absolute $c > 0$. The conclusion follows from (6), after adjusting the absolute constants. □

3.4. **Summation over denominator ranges.** It remains to sum the preceding estimates. The height bound (5) ensures that all exceptional denominator ranges occur at logarithmic heights $o(N)$. Consequently, the estimates of Lemma 3.6 apply in the tail of a convergent series.

Proof of Proposition 2.1. Fix $4 < \nu \le 5$ and choose $1 < \theta < \nu/4$. For $N \in \mathcal{D}$, put

$$w_j = N\theta^{-j}, \qquad x_j = \log(N/w_j) = j \log\theta \qquad (j = 0, 1, \ldots).$$

The intervals $(w_j/\theta, w_j]$ partition $(0, N]$. By (5), they cover the logarithms of all exceptional denominators $v > N^{1/3}$ once N is sufficiently large.

Suppose $\mathcal{A}(w_j) \neq \varnothing$. Some exceptional denominator in that range satisfies

$$\frac{1}{3}\log N < \log v \le w_j < \theta \log v \le \theta W_N.$$

In particular, $w_j \ge 1$ for sufficiently large N. Moreover, if $W_N > 0$, then

$$x_j > \log\frac{N}{\theta W_N} \longrightarrow \infty \qquad (N \to \infty,\ N \in \mathcal{D}),$$

by (5). Thus every nonempty range eventually satisfies the hypotheses of Lemma 3.6. If $W_N = 0$, there are no exceptional fractions to count.

For fixed $\theta > 1$, the series

$$\sum_{j\geq 0} \exp\left(-\frac{cx_j}{\log(2+x_j)}\right)$$

converges. Indeed, since $x_j = j\log\theta$, its terms are bounded by $\exp(-c_\theta\sqrt{j})$ for all sufficiently large j, with some $c_\theta > 0$.

Given any $J \geq 1$, every nonempty range has $j \geq J$ for all sufficiently large N. Lemma 3.6 therefore gives

$$\limsup_{\substack{N\to\infty \\ N\in\mathcal{D}}} \frac{1}{N}\sum_{j\geq 0} |\mathcal{A}(w_j)| \leq C\sum_{j\geq J} \exp\left(-\frac{cx_j}{\log(2+x_j)}\right),$$

where C is absolute. Letting $J \to \infty$ shows that

$$\sum_{j\geq 0} |\mathcal{A}(w_j)| = o_\nu(N).$$

Every index admitting an exceptional denominator greater than $N^{1/3}$ belongs to at least one of these sets. Consequently,

$$E_N(N^{1/3}, \nu) = o_\nu(N).$$

Finally, Lemma 3.1 gives

$$E_N(D, \nu) \leq C\big(ND^{2-\nu} + N^{2/3}\big) + o_\nu(N),$$

with an absolute constant C. Dividing by N and taking the upper limit along $\mathcal{D}$ proves the proposition. □

AI use declaration

The author used ChatGPT and Codex in developing the counting argument of Section 3, including the reduction to twofold products and the use of rational-power classes, and in formulating the diagonal construction in Section 2. These tools were also used to review the exposition and check proofs and references. The author takes full responsibility for the mathematical arguments and citations.

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